

# Math 55 Spring 2016 Practice Final Exam

## Solution Key

1. True or False:

- (a) **False.** The first statement allows  $z$  to depend on  $y$ , while the second requires one fixed  $z$  to work for all  $y$ . For instance, over a domain with at least two elements, take  $P(x, y, z, w)$  to mean  $z = y$ .
- (b) **False.** For example,  $\sqrt{2}$  and  $2 - \sqrt{2}$  are positive irrational numbers, but their sum is 2.
- (c) **False.** Take  $p = 3$ ,  $a = b = 2$ ,  $c = 1$ , and  $d = 4$ . Then  $a \equiv b \pmod{3}$  and  $c \equiv d \pmod{3}$ , but

$$2^1 \equiv 2 \pmod{3}, \quad 2^4 \equiv 1 \pmod{3}.$$

- (d) **False.** For each fixed finite vertex set there are only finitely many simple graphs, and the set of all finite vertex sets can be listed countably. Thus the set of all finite simple graphs is countable.
- (e) **False.** The events are disjoint, so they can be independent only if at least one has probability 0. For example, if  $X = 1$  with probability  $1/2$  and  $X = -1$  with probability  $1/2$ , then both events have probability  $1/2$ , but their intersection has probability 0.

2. **Disprove.** Among  $p, p + 2, p + 4$ , one number is divisible by 3. Since  $p > 3$  is prime,  $p \not\equiv 0 \pmod{3}$ . If  $p \equiv 1 \pmod{3}$ , then  $p + 2 \equiv 0 \pmod{3}$ , so  $p + 2$  is composite. If  $p \equiv 2 \pmod{3}$ , then  $p + 4 \equiv 0 \pmod{3}$ , so  $p + 4$  is composite. Therefore no such prime  $p > 3$  exists.

3. Write  $x = 2 + 5k$ . Then

$$2 + 5k \equiv 3 \pmod{9},$$

so  $5k \equiv 1 \pmod{9}$ . Since  $5^{-1} \equiv 2 \pmod{9}$ , we get  $k \equiv 2 \pmod{9}$ . Hence

$$x \equiv 2 + 5 \cdot 2 \equiv 12 \pmod{45}.$$

The full solution set is

$x = 12 + 45t, \quad t \in \mathbb{Z}.$

4. **Disprove.** The relation is reflexive and symmetric, but it need not be transitive. Let

$$x = 0, \quad y = p, \quad z = p + q.$$

Then  $xRy$ , since  $0 \equiv p \pmod{p}$ , and  $yRz$ , since  $p \equiv p+q \pmod{q}$ . But  $x \not R z$ , because  $p+q$  is divisible by neither  $p$  nor  $q$  when  $p$  and  $q$  are distinct primes. Thus  $R$  is not an equivalence relation.

5. Such a bijection is a permutation of  $\{1, 2, 3\}$  satisfying  $f^2 = \text{id}$ , so it is an involution. The possible cycle types are the identity permutation and a single transposition with the third element fixed. There is 1 identity permutation and  $\binom{3}{2} = 3$  transpositions. Therefore the number of such bijections is

$$\boxed{4}.$$

6. We prove this by induction. The base cases are  $a_0 = x$  and  $a_1 = y$ , which are rational by assumption. Now suppose  $a_{n-1}$  and  $a_{n-2}$  are rational. Since the rationals are closed under squaring and addition,

$$a_n = a_{n-1}^2 + a_{n-2}$$

is rational. Hence  $a_n$  is rational for every  $n \geq 0$ .

7. Let  $I_j$  be the indicator random variable for the event that bin  $j$  is empty.

- (a) Since a fixed bin is missed by each ball with probability  $3/4$ ,

$$\mathbb{E}[I_j] = \left(\frac{3}{4}\right)^{10}.$$

By linearity of expectation, the expected number of empty bins is

$$\boxed{4 \left(\frac{3}{4}\right)^{10}}.$$

- (b) By inclusion-exclusion, the probability that no bin is empty is

$$1 - 4 \left(\frac{3}{4}\right)^{10} + 6 \left(\frac{2}{4}\right)^{10} - 4 \left(\frac{1}{4}\right)^{10}.$$

Equivalently,

$$\boxed{\frac{4^{10} - 4 \cdot 3^{10} + 6 \cdot 2^{10} - 4}{4^{10}}}.$$

8. The first roll always produces the first face. After that, each roll has probability  $5/6$  of producing a new face, so the expected additional number of rolls until two different faces have appeared is  $1/(5/6) = 6/5$ . Once two faces have appeared, each roll has probability  $4/6 = 2/3$  of producing a third different face, so the expected additional number of rolls is  $1/(2/3) = 3/2$ . Therefore the expected total number of rolls is

$$\boxed{1 + \frac{6}{5} + \frac{3}{2} = \frac{37}{10}}.$$

9. Because the problem says “nonempty words,”  $a_0 = 0$ . Also

$$a_1 = 3, \quad a_2 = 8,$$

since the only forbidden two-letter word is  $XX$ .

For  $n \geq 3$ , a valid word of length  $n$  either ends in  $Y$  or  $Z$ , giving  $2a_{n-1}$  possibilities, or ends in  $X$ , in which case the previous letter must be  $Y$  or  $Z$ , giving  $2a_{n-2}$  possibilities. Thus

$$a_n = 2a_{n-1} + 2a_{n-2} \quad (n \geq 3).$$

Let  $G(x) = \sum_{n=0}^{\infty} a_n x^n$ . Then

$$G(x) = \frac{3x + 2x^2}{1 - 2x - 2x^2}.$$

10. This graph is the  $n$ -dimensional hypercube, whose vertices are subsets of  $A = \{1, \dots, n\}$ .
- (a) Each vertex  $S \subseteq A$  has degree  $n$ , since its neighbors are exactly the sets obtained by adding or removing one element of  $A$ .
  - (b) The graph is bipartite for every  $n$ . Partition the vertices according to whether  $|S|$  is even or odd. Every edge changes  $|S|$  by 1, so every edge goes between the two parts.
  - (c) Since the graph is connected and every vertex has degree  $n$ , it has an Euler circuit exactly when  $n$  is even. Thus, for  $n > 1$ ,

$$\boxed{G \text{ has an Euler circuit iff } n \text{ is even.}}$$

- (d) The graph is connected for every  $n$ . Given subsets  $S, T$ , move from  $S$  to  $T$  by toggling one element at a time in the symmetric difference  $S \Delta T$ .
11. The sum of the degrees is

$$2 + 2 + 2 + 1 + 1 + 1 + 1 = 10,$$

so the graph has  $10/2 = 5$  edges. A connected graph on 7 vertices must have at least  $7 - 1 = 6$  edges. Therefore any simple graph with this degree sequence must be disconnected.

12. The notation in the problem should read that  $C$  is a simple circuit containing the edge  $e_1$ . Let  $e_1 = \{u, v\}$ . Since  $e_1$  lies on a circuit, there is another path from  $u$  to  $v$  using the remaining edges of the circuit. Now take any two vertices  $a, b \in V$ . Because  $G$  is connected, there is an  $a$ -to- $b$  path in  $G$ . If this path does not use  $e_1$ , it is still present in  $G - e_1$ . If it does use  $e_1$ , replace that traversal of  $e_1$  by the other path around the circuit from  $u$  to  $v$ . This gives an  $a$ -to- $b$  walk in  $G - e_1$ , and hence an  $a$ -to- $b$  path in  $G - e_1$ . Therefore  $G - e_1$  is connected.